

Coding & Solution Implementation Report

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	5 Marks

Solution Implementation Overview:

This section documents the completeness of the implemented AI FitTrack API codebase, demonstrating clean code practices, working features, and adherence to requirements.

Solution Summary:

- **Repository:** FitTrack Backend API (Local deliverables included)
- **Programming Language:** JavaScript (Node.js ES6+ Async/Await)
- **Frameworks & Libraries:** Express.js (v4.19.2), Mongoose (v8.4.1), @google/genai (v2.0.0), jsonwebtoken, bcryptjs, morgan, dotenv
- **Key Features Implemented:**
 - User registration with bcrypt hashing (10 salts) & signed JWT access tokens
 - Full Workout CRUD operations with strict user ownership enforcement
 - Multi-criteria regex search across workoutName, category, and date
 - Google Gemini 1.5 Flash AI Workout Recommendation Engine
 - Google Gemini AI Fitness Progress Insights Analyzer
 - Centralized error handler and Postman test verification suite
- **Pending Features:** None. 100% of planned functional and AI capabilities are implemented and tested.
- **Setup / Run Instructions:**
 1. Navigate to codebase directory & run npm install
 2. Configure .env with PORT=5000, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY
 3. Execute npm run dev or npm start
 4. Import FitTrack.postman_collection.json in Postman to test all endpoints

Coding & Solution — Quality Checklist & Code Verification

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Code Quality Checklist:

S.No	Criteria	Status	Verification Evidence in AI FitTrack API
1	Modular Directory Layout	Pass	Clear separation into config/, controllers/, middleware/, models/, routes/, services/.
2	Coding & Naming Standards	Pass	camelCase for methods and variables, PascalCase for Mongoose models (User, Workout).
3	Credential Isolation	Pass	All API keys and database strings loaded from .env; zero hardcoded secrets.
4	Standardized HTTP Statuses	Pass	Returns 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found.
5	Cryptographic Security	Pass	Bcrypt hashing with 10 salt rounds and signed HMAC-SHA256 JWT tokens.

Endpoint Verification Matrix:

Method	Endpoint Route	Operational Scope	Auth	Status	Result
POST	/api/auth/register	Register new user account	No	201 Created	Verified
POST	/api/auth/login	Authenticate & generate JWT	No	200 OK	Verified
GET	/api/auth/profile	Fetch user profile details	Bearer	200 OK	Verified
POST	/api/workouts	Log new workout activity	Bearer	201 Created	Verified
GET	/api/workouts	Fetch all user workouts (date sort)	Bearer	200 OK	Verified
GET	/api/workouts/:id	Fetch specific workout by ID	Bearer	200 OK	Verified
PUT	/api/workouts/:id	Update workout properties	Bearer	200 OK	Verified
DELETE	/api/workouts/:id	Delete workout entry	Bearer	200 OK	Verified
GET	/api/workouts/search?q=	Multi-criteria regex search	Bearer	200 OK	Verified
POST	/api/ai/workout-recommendation	Generate tailored workout routine	Bearer	200 OK	Verified
POST	/api/ai/fitness-insights	Evaluate workout progress trends	Bearer	200 OK	Verified